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Spectrometer Command Set for BTC110/BRC100/BRC110/BTC200

Default communication protocol: RS232

Required Settings

Baud rate: 9600 (BTC110, BRC100), 19200(BRC110)

Data bits: 8

Parity: None

Stop bits: 1

Flow control: None

Command Summary

Letter	Description
A	Set average number
I	Set integration time
B	Reserved
K	Set communication baud rate
Q	Initialize Spectrometer (parameter with default)
S	Start spectral acquisition
C	Set Coefficient to Flash
a	Set ASCII data mode
b	Set binary data mode (default)
?	Query operation parameters
P	Set Pixel mode (Only BRC110)
T	Set time delay
x	Set Multiplier (Only BTC200)

Command descriptions



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Set Spectrometer to be initialized

Description: Sets all operation parameters to their default value.

Command syntax:	Q
Response:	ACK or NAK
Examples:	Q

Default value:

For BTC110: baud=9600,inttime=21ms,avenum=1,data mode=binary compression mode.

For BRC100: baud=9600,inttime=35ms,avenum=1,data mode=binary compression mode.

Set ASCII data mode

Description: Sets the mode in which data values are interpreted to be ASCII. Only unsigned integer strings (For BTC110:0-65535, For BRC100:0-4096) are allowed in this mode. Data values are terminated with a carriage return (ASCII 13) or linefeed (ASCII 10).

Command syntax:	A
Response:	ACK or NAK
Examples:	A

Set Binary data mode

Description: Sets the mode in which data values are interpreted as binary. In this mode, For BTC110, 16 bit unsigned integer values (0-65535)
For BRC100, 12 bit unsigned integer values (0-4096)

--with the MSB followed by the LSB--are allowed. (Format: first is MSB then is LSB)

Command syntax:	B
Response:	ACK or NAK
Examples:	B

Data mode (default) is compressed binary data mode. Please see notes.

Notes: BTC110/BRC100 data compression

Transmission of spectral data over the serial port is relatively slow process. Even at 115200 baud. The transmission of a complete 2048-point spectrum takes around 400 ms. The BTC110/BRC100 implements a data compression routine that minimizes the amount of data that needs to be transferred over the RS-232 connection. Using command "b" enables the data compression.

Every scan transmitted by the BRC100 will then be compressed. The compression algorithm is as follows:

1. The first pixel data (a 16-bit unsigned integer) is compared to
 - If the data different ≥ 128 , Then send 3 Byte as follow

0x80

Data High

Data Low

- If the data different <128 , Then send 1 Byte as follow

Data Different

2. The next pixel data is compared to previous pixel data.

- If the data different ≥ 128 , Then send 3 Byte as follow

0x80

Data High

Data Low

- If the data different <128 , Then send 1 Byte as follow

Data Different

3. Repeat step 2 until all pixels have been read.

The following shows a section of a spectral line source spectrum and the results of the data compression algorithm.

Pixel value	Value difference	Transmitted bytes
185	0	0x80 0x00 0xb9
2151	1966	0x80 0x08 0x67
836	-1315	0x80 0x03 0x44
453	-383	0x80 0x01 0xc5
210	-243	0x80 0x00 0xd2
118	-92	0xa4
90	-28	0xe4
89	-1	0xff
87	-2	0xfe
89	2	0x02
86	-3	0xfd
88	2	0x02
98	10	0x0a
121	23	0x17
383	262	0x80 0x01 0x7f
1162	779	0x80 0x04 0x8a
634	-528	0x80 0x02 0x7a
356	-278	0x80 0x01 0x64
211	-145	0x80 0x00 0xd3



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132	-79	0xb1
88	-44	0xd4
83	-5	0xfb
86	3	0x03
82	-4	0xfc
91	9	0x09
92	1	0x01
81	-11	0xf5
80	-1	0xff
84	4	0x04
84	0	0x00
85	1	0x01
83	-2	0xfe
80	-3	0xfd
80	0	0x00
88	8	0x08
94	6	0x06
90	-4	0xfc
103	13	0x0d
111	8	0x08
138	27	0x1b

In this example, spectral data for 40 pixels is transmitted using only 60 bytes. If the same data set were transmitted using uncompressed data, it would require 80 bytes.



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Spectral Acquisition

Command syntax:	S
Response:	If successful, 2048 pixels data (data compression mode, so maybe not 2048 byte)
Examples:	S

Set average number

Description: Sets the number of discrete spectra to be averaged.

Command syntax:	A {DATA WORD}
Response:	ACK or NAK
Range:	1-15 (For BRC100) 1-65535 (For BRC110, BTC110)
Default value:	1
Examples:	A 1

Set Integration time

Description: Sets the BRC100 integration time, in milliseconds, to the value specified.

Command syntax:	I {DATA WORD}
Response:	ACK or NAK
Range:	21-65000 (for BTC110), 35-65000 (For BRC100, BRC110)
Default value:	21 (For BTC110), 35 (For BRC100, BRC110)
Examples:	I 21

Set baud rate

Description: Sets the BRC100's baud rate.

Command syntax:	K {DATA WORD}
Response:	ACK or NAK
Value:	7=600 6=1200 5=2400 4=4800 3=9600 2=19200 1=38400 0=115200
Default value:	3 (BTC110, BRC100), 2 (BRC110)
Example:	K 2

Set Coefficients

Description: Set coefficients.

Wavelength=coef_0+coef_1*pixel+coef_2*pixel*pixel+coef_3*pixel*pixel*pixel

Command syntax:	C n {DATA WORD}
Response:	ACK or NAK



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Range:	N=1-4, data word=float number
Examples:	C 1 1.23E-5 ----- coef_0=1.23e-5 C 2 2.345e+2 ----- coef_1=2.34e+2 C 3 123 -----coef_2=12 C 4 12345E-10 -----coef_3=12345e-10

Query parameter

Description: Returns the current value of the parameter specified. The syntax of this command requires two ASCII character. The second ASCII character corresponds to the command character that sets the parameter of interest (acceptable values are A, K, I, a).

Command syntax:	? {ASCII character}
Response:	ACK followed by {DATA WORD} Response value :(16 bit)
Examples:	?A : Query average number ?K : Query baud rate ?I : Query integration time ?a : Query data mode ?C : Query coefficients ?v : Query spectrometer version ?t : Query cooler temperature (Only For BTC110) ?B : reserved ?P : Query pixel mode (only for BRC110) ?T : Query time delay ?x : Query Multiplier (only for BTC200)

Notes: Specialty

Response of command (?C):

Char0-Char 12	Char13-Char 24	Char25-Char 36	Char 37-Char48	10	13
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Char0 ---Char12: *.***E** --Coefficient a0
 Char13---Char24: *.***E** --Coefficient a1
 Char25---Char36: *.***E** --Coefficient a2
 Char37---Char48: *.***E** --Coefficient a3



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Set pixel mode (only for BRC110)

Description: Specifies which pixels are transmitted while all pixels are acquired on every scan; this parameter determines which pixels will be transmitted.

Command syntax:	P {DATA WORD}
Response:	ACK or NAK
Value:	0 =all 2048 pixel s 1=every n (th) pixel with no averaging. 2=every n (th) pixel with averaging (effective boxcar width of n , by another word, the n (th) value should be less than n of B command) 3=pixel x through y every n pixels 4=up to 6 randomly selected pixels between 0 and 2047 (denoted p1, p2, p6)
Default value:	0
Examples:	P 0 (P 0) P 1 10 (P 1 n) P 2 20 (P 2 n) P 3 400 700 10 (P 3 x y n) P 4 6 100 200 300 400 500 600 (P 4 n p1 p2 ...pn)

Set time delay (only for BRC110)

Description: Sets the BRC100 time interval, in milliseconds, to the value specified.

On pixel mode 4, scan 256 times, between every scan is the time delay.

Command syntax:	T {DATA WORD}
Response:	ACK or NAK
Range:	35-65000
Default value:	35
Examples:	T 35

Examples

Example 1

a	(Set ASCII data mode)
Q	(Reset Spectrometer to default conditions)
I 200	(Set integration time to 200ms)
S	(Acquire spectrum)